



Faculty of engineering, Cairo University
Electronics and Electrical Communication Department
ELC4046 – Process Control



Lab 2 _ PID CONTROLLER CHARACTERISTICS

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Submitted to :

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LABORATORY EXERCISE 8

PID CONTROLLER CHARACTERISTICS

OBJECTIVE: To demonstrate the characteristics of the proportional, integral (reset) and derivative control modes in open and closed loop operations, including definitions of the tuning parameters.

PREREQUISITE: Completion of Laboratory Exercise 0 Getting Familiar or an equivalent amount of familiarity with the program operation.

1. RUNNING THE PROGRAM

Run **PC-ControLAB**.

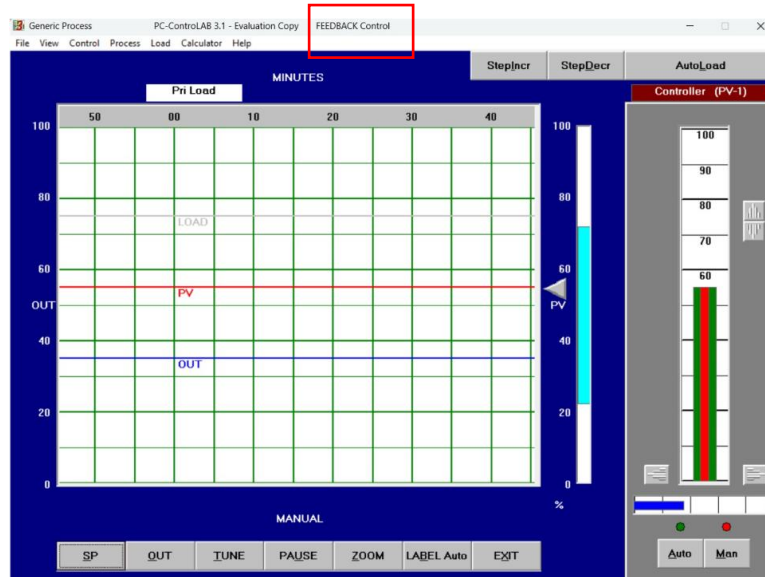
Slow down the simulation speed via **View | Scroll Rate** and select a mid-value.



2. PROPORTIONAL MODE DEMONSTRATION

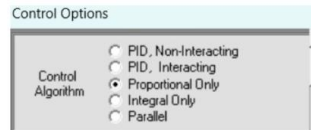
2.1 Set Up

Check the top line of the display. Be sure that you are using the GENERIC process model, the FEEDBACK control strategy. Be sure that the controller is in MAN.





From the Menu Bar, select **Control | Control Options**.
For Control Algorithm, select **Proportional Only**.



While you still have the Control Options box on display, be sure the following options are selected:

Control Action

REVERSE

Set Pt Tracking

NO

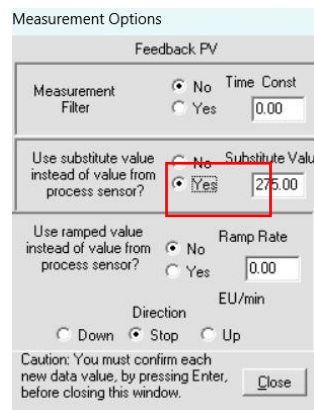


Press **Close** to remove the Control Options box.

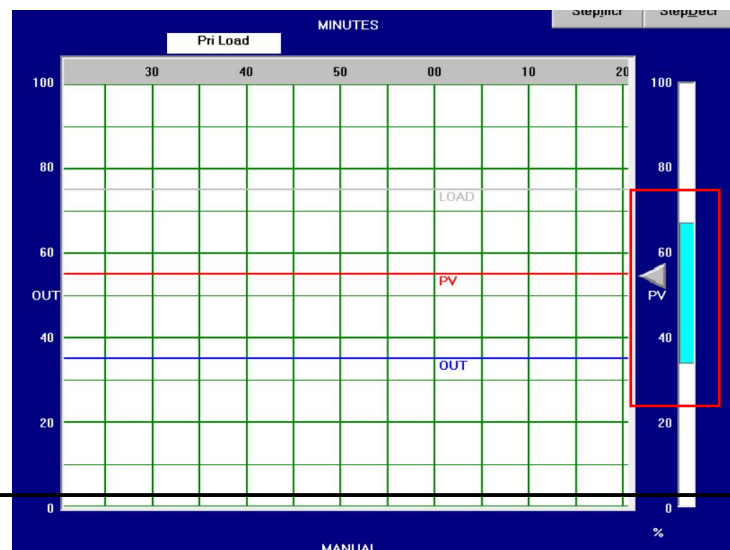
Select **Control | Measurement Options**.

Select **Yes** for “Use substitute value instead of value from process sensor?”

Press **Close** to remove the Measurements Option box.



Select **View | Display Range | Percent of Span**.
Select **View | Display Proportional Band**. (An auxiliary bar display will appear between the strip chart and the instrument faceplate. Its use will be explained later.)



2.2 Gain and Proportional Band Features

Press **TUNE**. Set or confirm the following values:

GAIN

2.0

MANUAL RESET

35

TUNING PARAMETERS

Proportional Only Controller

Tuning Options Schedule

Tuning Parameters

	Present	New	
Gain	2.00		OK
Man Reset Percent	35.00		OK

Click on the **Options** tab and select “Display proportional tuning parameter as” **Prop Band**.

Tuning Options Schedule

Tuning Parameter Options

Display Proportional Tuning Parameter as:	Display Reset Tuning Parameter as:
☐ Gain	☒ Min/Rpt ☐ Sec/Rpt
☒ Prop Band	☐ Rpt/Min ☐ Rpt/Sec
Reset Action	Derivative Gain
☒ ON ☐ OFF	Present New

Return to the **Tuning** tab.

What value and what name does the Tuning Display now show in place of GAIN?

Value 50% Name Prop Band Percent

TUNING PARAMETERS

Proportional Only Controller

Tuning Options Schedule

Tuning Parameters

	Present	New	
Prop Band Percent	50		OK
Man Reset Percent	35.00		OK

Recall that: Prop Band = $\frac{100}{\text{Gain}}$ = **100/2=50%**

Press **Close** to remove the Tuning dialog box.

Select **View | Data Monitor**.

Record the following present values

Operating Data	
PV	55.00
SP	55.00
Out	35.00
Close	

Process variable: 55

Setpoint: 55

Controller output: 35

For a Reverse Acting controller, the controller error is calculated as:

$$\text{ERROR} = \text{SP} - \text{PV}$$

and the top and bottom of the proportional band can be calculated from the present values of setpoint (SP), manual reset (MR) and gain (or PB).

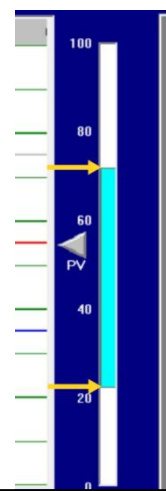
$$\text{PB}_{\text{top}} = \text{SP} + \text{MR} \times (\text{PB} / 100)$$

$$\text{PB}_{\text{bot}} = \text{PB}_{\text{top}} - \text{PB}$$

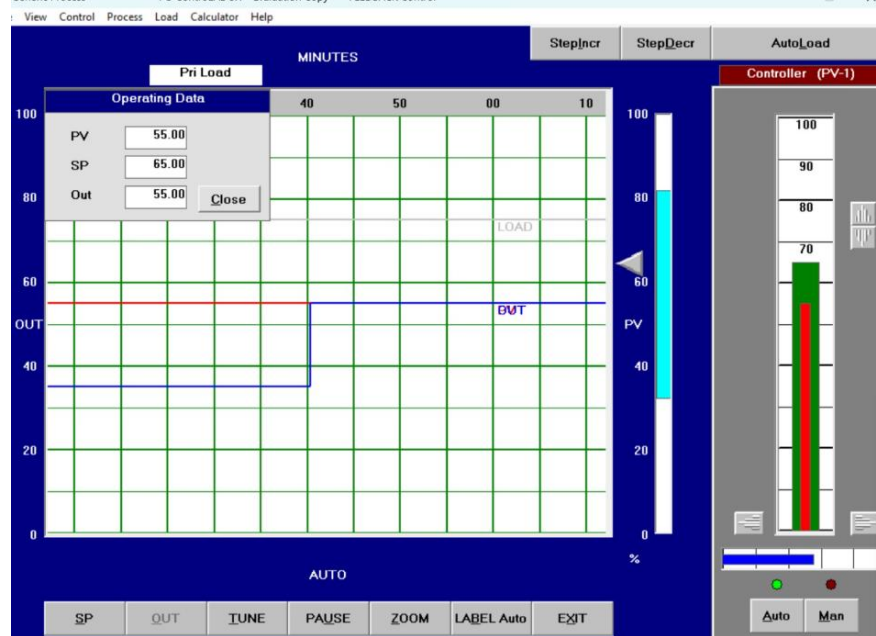
Calculated: 72.5% Observed: 72.5%

Calculated: 22.5% Observed: 22.5%

PB_{top}
PB_{bot}



Put the controller in AUTO. Change the Set Point to 65.



Did the PB bar move as expected?

yes, it has shifted upwards

The **proportional band (PB)** keeps the same width, but when the **setpoint (SP)** increases, both **PB_{top}** and **PB_{bot}** increase, so the entire PB shifts upward

Calculate the theoretical controller output from:

$$\text{OUTPUT} = \text{GAIN} * \text{ERROR} + \text{MR} = (2 * (65 - 55) + 35) = 55 \%$$

55%

Does the theoretical output agree with the observed output?

yes

From an observation of the Proportional Band bar, what value of PV would cause the controller output to go:

Substitute in the equation : **PV = SP - ((OUT - MR) / GAIN)**
to zero?

82.5%

to 100%?

32.5%

(Recall that the Proportional Band is the range through which the PV must travel to cause the controller output to change by 100%. For a reverse acting controller, as the PV drops from PB_{top} to PB_{bot}, the controller output goes from 0% to 100%.)

Select **Control | Measurement Options**. Enter a substitute PV value of 82.5.

Measurement Options

Feedback PV

Measurement Filter ☐ No ☐ Yes Time Const 0.00

Use substitute value instead of value from process sensor? ☐ No ☒ Yes Substitute Value 82.5

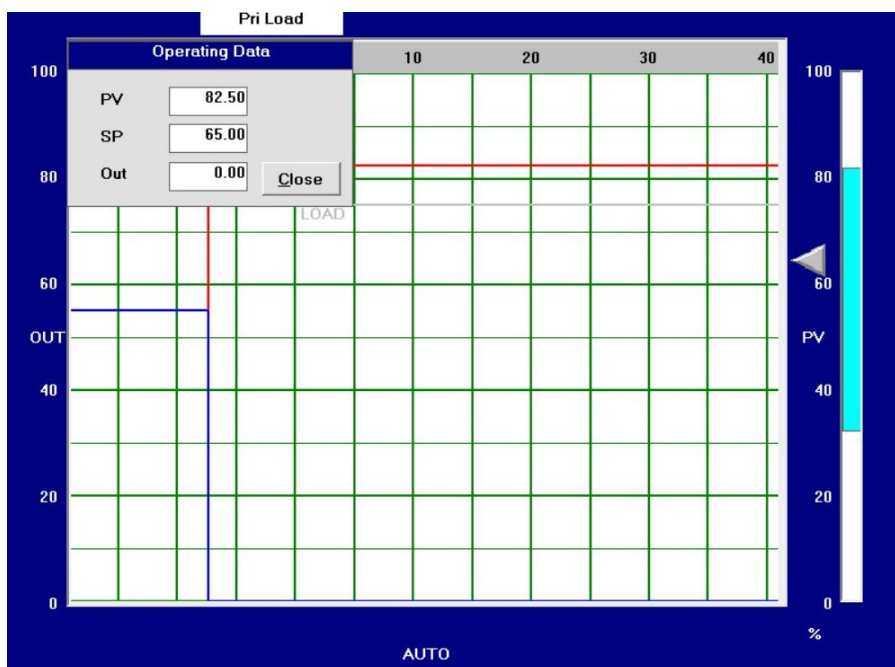
Use ramped value instead of value from process sensor? ☐ No ☐ Yes Ramp Rate 0.00

Direction EU/min

Caution: You must confirm each new data value, by pressing Enter, before closing this window.

Close

Controller Output: zero



(At this point it is difficult to tell whether the controller output has been calculated at 0.0, or whether it is actually some lower value, but "pegged out" at 0.0. Therefore, back off on the PV slightly by entering a substitute value of 82. That should produce a controller output just slightly above 0.0.)

Now enter a substitute PV value of 32.5.

Controller Output: 100%

Measurement Options

Feedback PV

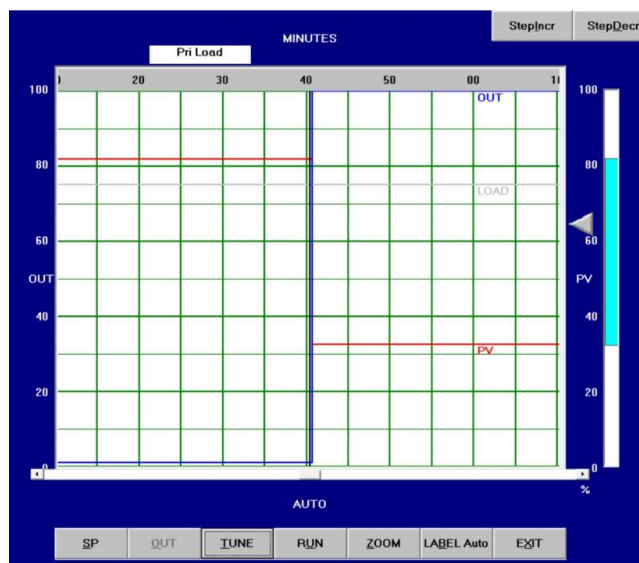
Measurement Filter ☒ No ☐ Yes Time Const 0.00

Use substitute value instead of value from process sensor? ☒ No ☐ Yes Substitute Value 32.5

Use ramped value instead of value from process sensor? ☒ No ☐ Yes Ramp Rate 0.00

Direction ☐ Down ☒ Stop ☐ Up EU/min

Caution: You must confirm each new data value, by pressing Enter, before closing this window.



(You can raise the substitute PV value slightly, say to 33, to demonstrate that the output value was truly at 100.0, not just pegged out at the top of the chart.)

2.3 Closed Loop Proportional Mode Response

Put the controller in MAN. Change the Set Point to 55. Change the controller output to 35.0. Change the substitute PV value to 55. Then select **Control | Measurement Options** and select **NO** for “Use substitute value ...?” (Now we will be using process feedback.) Now put the controller in AUTO.

Feedback PV

Measurement Filter	☒ No Time Const ☐ Yes 0.00
Use substitute value instead of value from process sensor?	☒ No Actual Value ☐ Yes 55.00
Use ramped value instead of value from process sensor?	☒ No Ramp Rate ☐ Yes 0.00
Direction ☐ Down ☒ Stop ☐ Up	
Caution: You must confirm each new data value, by pressing Enter, before closing this window.	
Close	

Change the Set Point to 65. When the control loop comes to equilibrium, note:

Operating Data

PV	55.00	SetPoint Ramp?	☒ No
SP	55.00		☐ Yes
Out	35.00		
New Set Point	65	OK	Close

PV

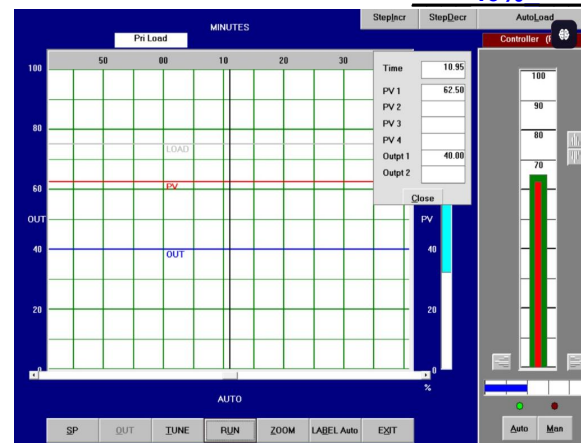
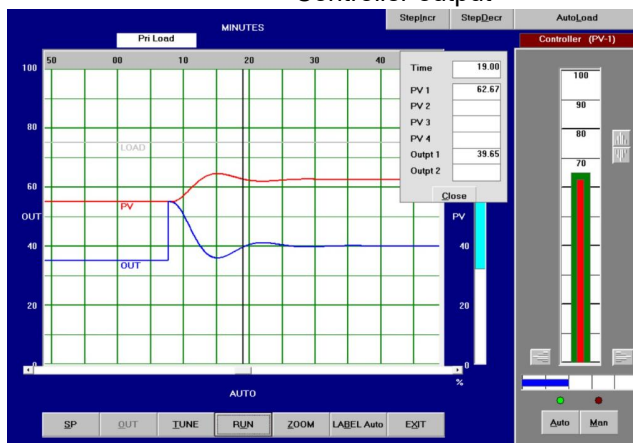
62.5%

SP

65

Controller output

40%



Calculate the error, then calculate the theoretical controller output from the equation given previously. Does this agree with the observed output? yes

$$\text{ERROR} = \text{SP} - \text{PV} = 65\% - 62.5\% = 2.5\%$$

$$\begin{aligned} \text{OUTPUT} &= \text{GAIN} * \text{ERROR} + \text{MR} \\ &= 2.0 * 2.5\% + 35\% = 40.0\% \end{aligned}$$

The PV and SP do not agree. Which way, increase or decrease, should the Manual Reset be adjusted to bring the PV into agreement with the SP?

For the PV to agree with SP The manual reset should be increased because the error is positive.

Experimentally adjust the Manual Reset until the PV reaches the SP.

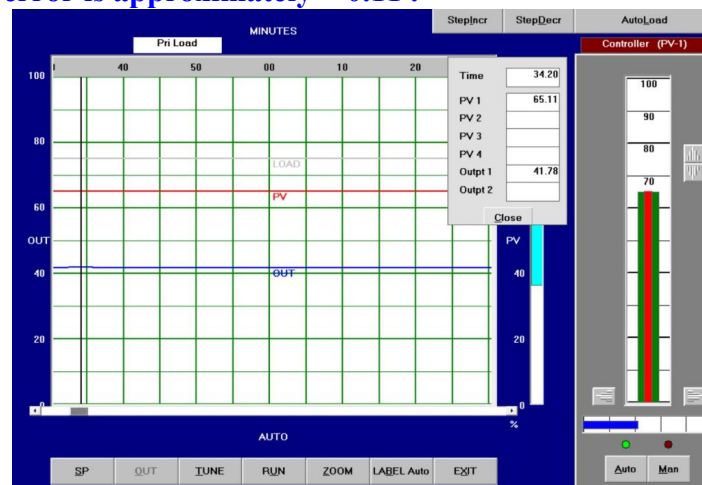
After some trials, the suitable value of **MR** that makes **PV** equal to **SP** was found to be **42**

Final value of Manual Reset:

	Present	New	
Prop Band Percent	50		OK
Man Reset Percent	42.00		OK

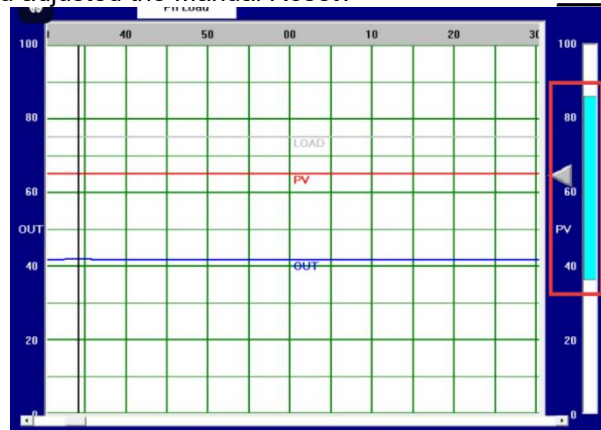
Final value of Controller output:

The controller output reaches approximately 41.78, corresponding to a process variable (PV) of 65.11, where the error is approximately -0.11 .



Did the PB bar move as you adjusted the Manual Reset?

yes, it shifted upward



2.4 Controller Action (Direction) Demonstration

Put the controller in MAN. Select **Control | Control Options**.

For Control Action, select **DIRECT**.

Press **Close** to remove the Control Options box.

Now put the controller back in AUTO.

Control Options	
Control Algorithm	☐ PID, Non-Interacting
	☐ PID, Interacting
	☒ Proportional Only
	☐ Integral Only
	☐ Parallel
Control Action	☒ Direct ☐ Reverse
Set Point Tracking	☒ No ☐ Yes

Wait for a while. What do you observe?

The process variable (PV) decreases over time, and the controller output (M) also decreases until both reach zero. This occurs because the controller is direct-acting, where a decrease in PV leads to a decrease in the controller output. This creates a positive feedback effect, driving the system further away from the setpoint instead of stabilizing it.



What you should have observed:

1. The relationship between controller Gain and Proportional Band width.
2. The relationship between the Set Point, Manual Reset and Proportional Band position.
3. The relationship between Proportional Band position, PV and controller output, when the controller is in AUTO.
4. That Proportional-Only control will not (usually) cause the PV and SP to agree, unless Manual Reset is adjusted.
5. That adjusting the Manual Reset (in AUTO) is equivalent to shifting the Proportional Band.
6. Wrong Controller Action setting gives a disastrous unstable feedback loop.

3. INTEGRAL MODE DEMONSTRATION

3.1 Set Up

Select **View** and click on **Display Proportional Band** (to remove it).

Put the controller in MAN.

Select **Control | Control Options** and set or check the following options:

Control Algorithm:

Action, Direct or Reverse:

Set Point Tracking?

Deriv on Error or Meas:

Prop on Error or Meas:

PID, Non-Interacting

Reverse

No

Meas

Error

Control Options

Control Algorithm	☒ PID, Non-Interacting ☐ PID, Interacting ☐ Proportional Only ☐ Integral Only ☐ Parallel
Control Action	☐ Direct ☒ Reverse
Set Point Tracking	☒ No ☐ Yes
Remote Setpoint Source	☒ No ☐ Yes ☐ PV-1 ☐ PV-2 ☐ PV-3 ☐ PV-4
Deriv On Error or Meas	☐ Error ☒ Meas
Prop On Error or Meas	☒ Error ☐ Meas
Reset Feedback Source	☒ No ☐ Yes ☐ PV-1 ☐ PV-2 ☐ PV-3 ☐ PV-4

Go to **Control | Measurement Options** and set:

Use substitute value:

Substitute measurement value:

YES

50

Measurement Options

Feedback PV	
Measurement Filter	☒ No ☐ Yes
Use substitute value instead of value from process sensor?	☐ No ☒ Yes
Use ramped value instead of value from process sensor?	☐ No ☐ Yes
Ramp Rate	0.00
Direction	☐ Down ☒ Stop ☐ Up

Press **Tune | Options** and set the following:

Display the proportional tuning parameter as

Return to the **Tuning** tab and set

Gain: 1.0

Reset, Min/Repeat: 5.0

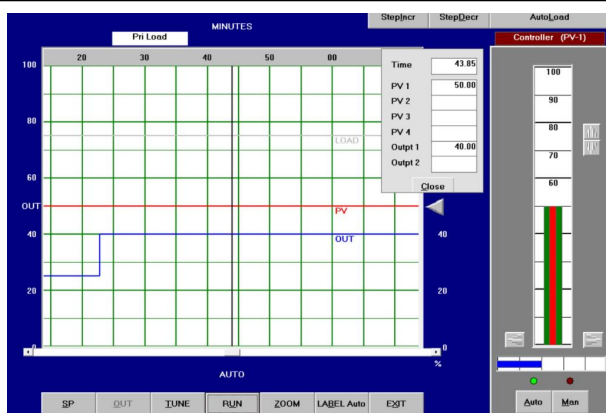
Deriv, Mins: 0.0

Set the following:

Set Point: 50

Controller Output: 40

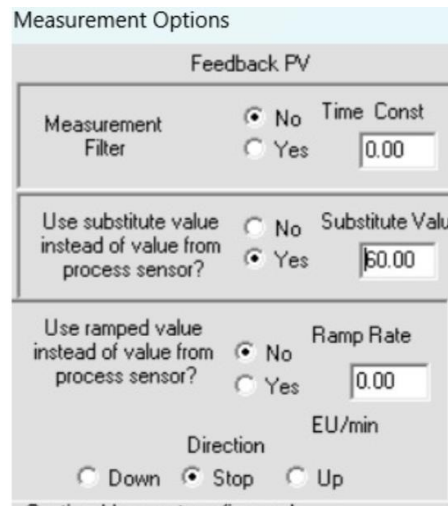
Change to AUTO. (You should now see the control loop simulation in stable operation with measurement and setpoint both at 50% and the valve signal at 40%.)



Integral Mode Response

Go to **Control | Measurement Options** and enter:

Substitute measurement value 60



Measurement Options

Feedback PV

Measurement Filter: ☒ No ☐ Yes Time Const: 0.00

Use substitute value instead of value from process sensor?: ☐ No ☒ Yes Substitute Value: 60.00

Use ramped value instead of value from process sensor?: ☒ No ☐ Yes Ramp Rate: 0.00

Direction: ☐ Down ☒ Stop ☐ Up

EU/min

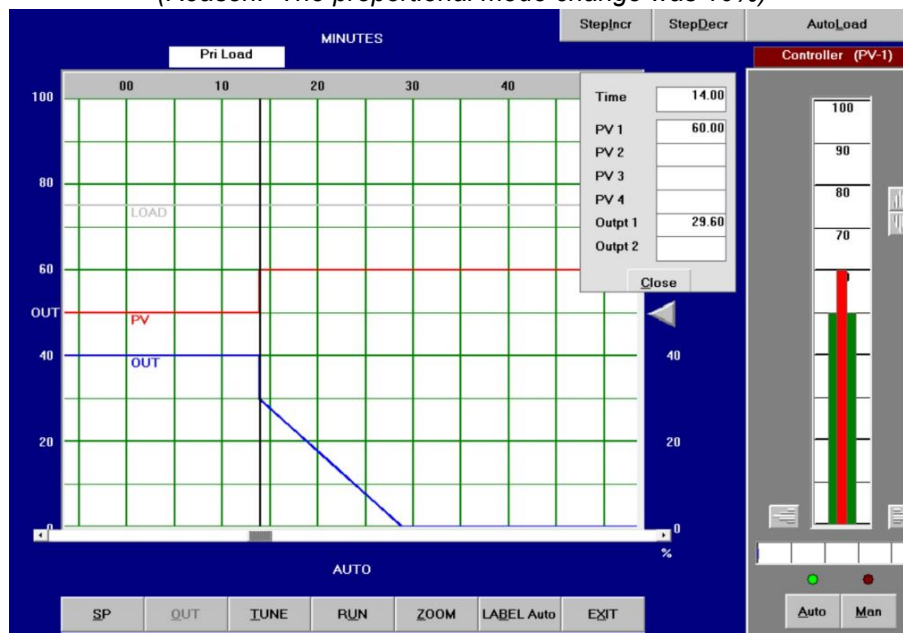
Observe:

The controller output signal immediately drops from 40% to 30%.

(Reason: The controller is reverse acting, so a measurement increase causes a controller output decrease. The GAIN is 1.00, so the change in controller output due to proportional action is 1 times the measurement change.)

The controller output then begins to ramp downward, at the rate of 2% per minute.

(Reason: The proportional mode change was 10%)



What you should have observed:

A 2%/minute change is equivalent to 5 minutes per 10% change, or 5 minutes per repeat of the proportional response.

Note that the ramping action continues until the output reaches a saturation limit (0%). This condition is called "reset windup."

3.3 Integral Mode Response, A Second Example

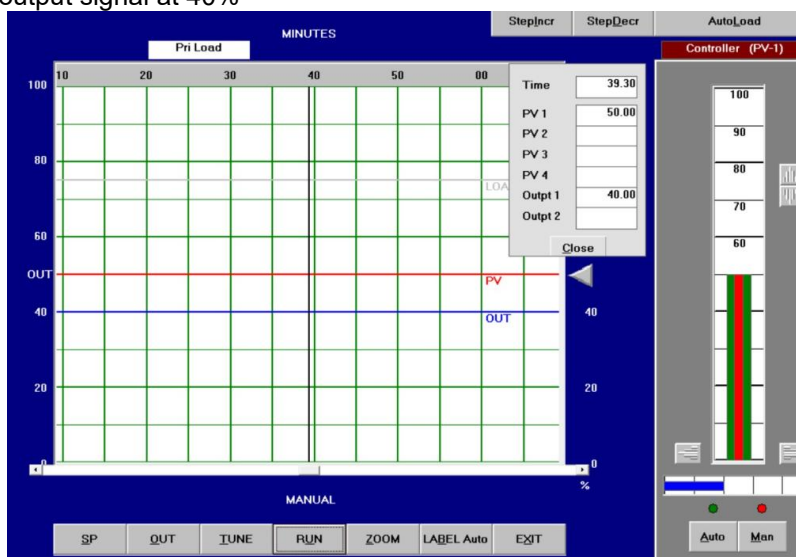
Put the controller in MAN.

Enter a substitute measurement value of 50.

Enter a controller output of 40.

Put the controller in AUTO.

Be sure the controller is in a steady state condition, with the measurement at 50% and controller output signal at 40%



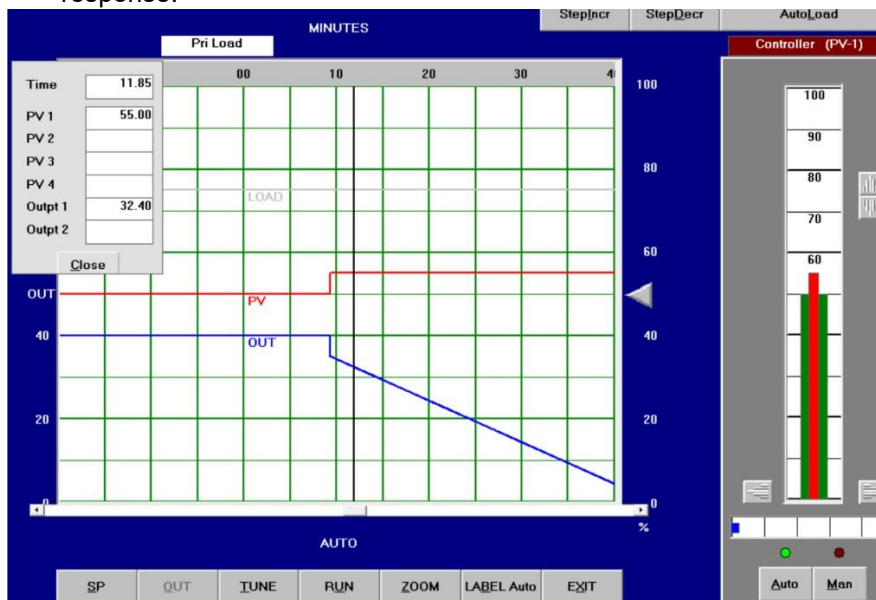
Now, enter a substitute measurement value of 55%.

Use substitute value instead of value from process sensor? ☐ No ☒ Yes Substitute Value

Observe:

The proportional response is now only 5%.

The controller output ramps downward at 1%/minute. This rate is equivalent to 5 minutes for 5% change, or again, it is 5 minutes per repeat of the proportional response.



4. DERIVATE MODE DEMONSTRATION

4.1 Set Up

Put the controller in MAN.

In the **Tune | Options** tab, set the following:

Reset Action	OFF
Derivative Gain	100.0

Enter the following tuning values:

Gain:	0.5
Reset, Min/Repeat:	Not applicable (it's OFF)

The first screenshot shows the 'NonInteractive PID' interface with the 'Options' tab selected. Under 'Tuning Parameter Options', 'Display Proportional Tuning Parameter as:' has 'Gain' selected. 'Display Reset Tuning Parameter as:' has 'Min/Rpt' selected. 'Reset Action' is set to 'OFF'. 'Derivative Gain' is set to 'Present: 100.00' and 'New' is empty.

The second screenshot shows the 'NonInteractive PID' interface with the 'Tuning Parameters' tab selected. It shows three parameters: 'Gain' (Present: 0.50, New: empty), 'Reset Min/Rpt (Reset)' (Present: 5.00, New: empty), and 'Deriv Minutes' (Present: 0.00, New: empty). Each parameter has an 'OK' button next to the 'New' field.

Deriv, Min:	0.0
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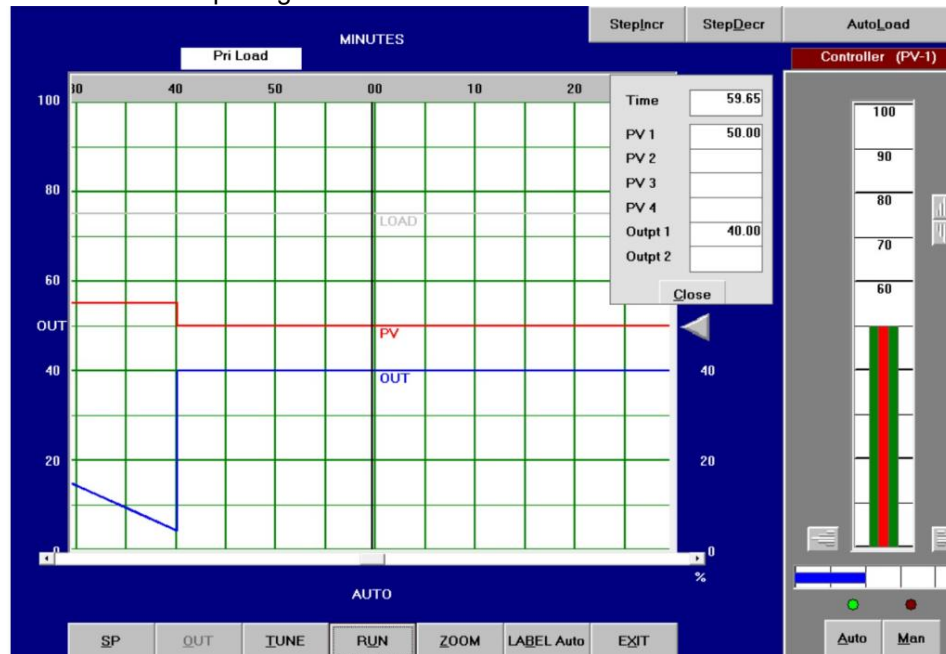
Enter the following operational values:

Set Point: 50
Controller output: 40

In the **Control | Measurement Options**, select **Yes** for “Use substitute value ...?”, then enter a substitute measurement value of 50

Change the controller mode to AUTO.

Verify that the controller is in stable operation with setpoint and process variable at 50, and controller output signal at 40.



4.2 Ramp Response with Proportional Mode Only

Under **Control | Measurement Options**, select **Yes** for “Use ramped value instead of value from process sensor?”

Enter a ramp rate of 4%/minute.

Select **Up**.

Use ramped value instead of value from process sensor? ☐ No ☒ Yes

Ramp Rate EU/min

Direction ☐ Down ☐ Stop ☒ Up

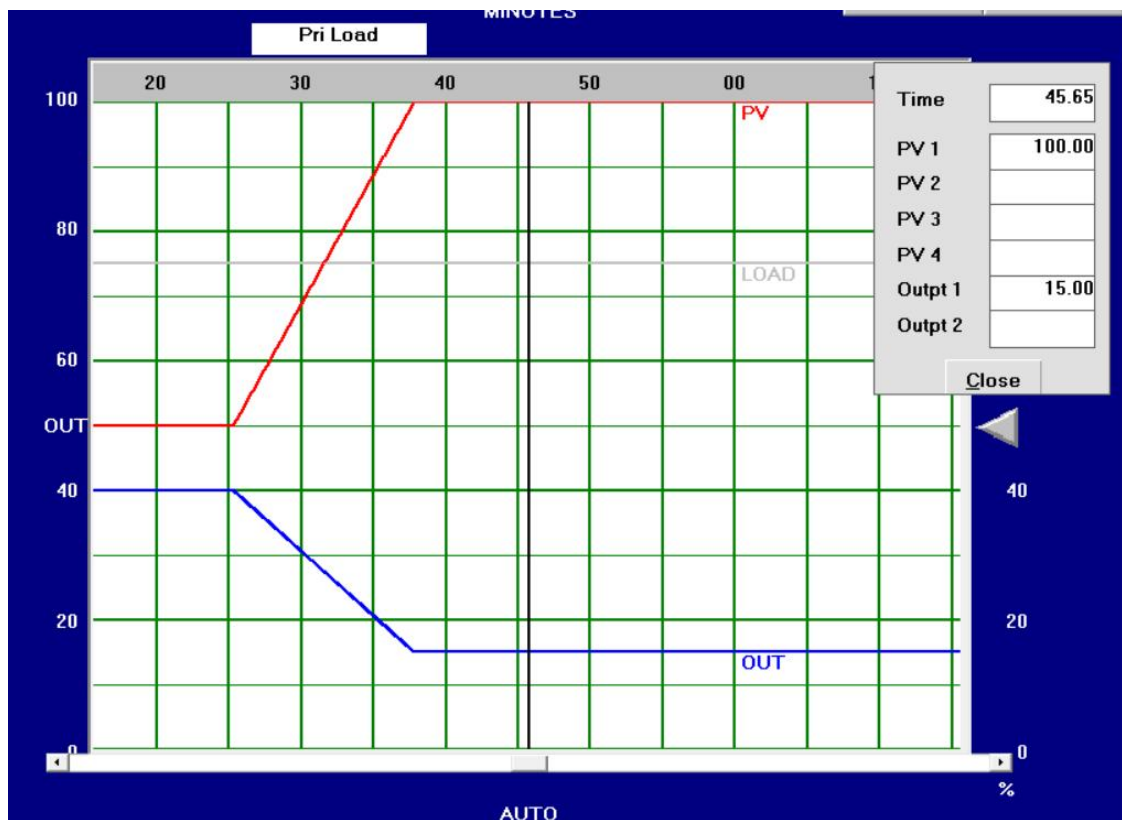
Caution: You must confirm each

Observe:

The process variable ramping upward from its initial value of 50% at the rate of 4%/minute.

The controller output ramping downward from its initial value of 40% at 2%/minute, stopping at 15% when the measurement reaches 100%.

(Reason: It is a reverse acting controller, so as measurement increases, the controller output decreases. The Gain is 0.5, so the controller output drops half as fast as the measurement rises.)



4.3 Ramp Response with Proportional and Derivative Modes

In **Control | Measurement Options** select **No** for “Use ramped value ...?”

Change the controller mode to MAN.

Enter a controller output signal of

40

In **Control | Measurements Options** enter a substitute measurement value of

50

Change the controller mode back to AUTO.

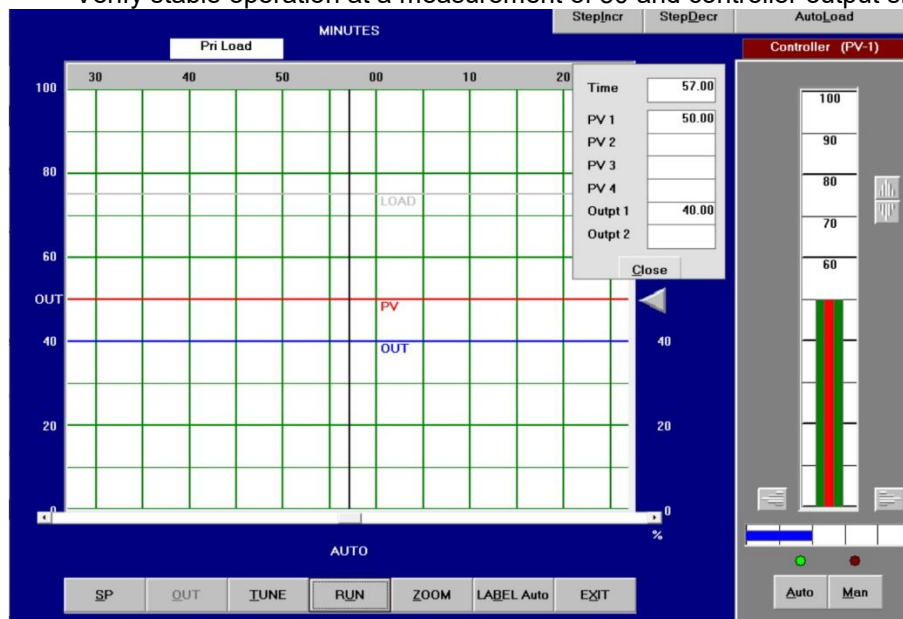
Press **Tune** and change the Deriv setting to 5 minutes.

The screenshot shows a 'Tuning Parameters' dialog box with three rows of controls. Each row has a label, a 'Present' value, a 'New' input field, and an 'OK' button. The first row is for 'Gain' with a present value of 0.50. The second row is for 'Reset' with a present value of 5.00 and sub-labels 'Min/Rpt' and 'Reset'. The third row is for 'Deriv' with a present value of 5.00 and a sub-label 'Minutes'.

	Present	New	
Gain	0.50		OK
Reset Min/Rpt Reset	5.00		OK
Deriv Minutes	5.00		OK

Verify stable operation at a measurement of 50 and controller output signal of

40



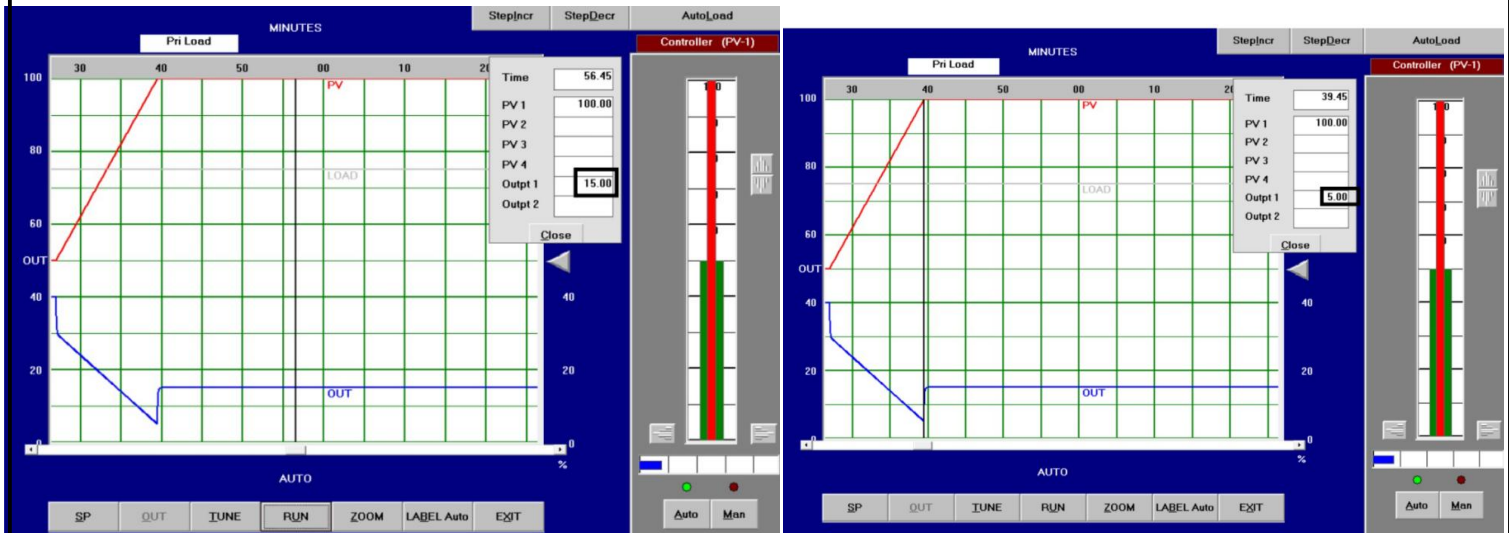
In **Control | Measurement Options**, enter the same ramp parameters as before:

Rate:
 Direction:
 then select **Yes** for Use ramped value...?"

4%/minute
 UP

Observe:

- The measurement ramping as before.
- The controller output signal makes an almost immediate change from 40% to 30% then ramps downward at the rate of 2% per minute to 5%, when the measurement has reached 100%. As soon as the measurement ramp stops, the controller output signal makes an almost immediate change from 5% to 15%, then remains constant.



Reason: The derivative contribution is

$$\begin{aligned}
 &= - \text{Gain} \times \text{Deriv time} \times \text{Meas ramp rate} \\
 &= - 0.5 \times 5 \text{ minutes} \times 4 \text{ \%/minute} \\
 &= 10 \%
 \end{aligned}$$

If the controller had no derivative (as in the first trial), you can estimate approximately how long, from the initiation of the ramp, was required for the controller output signal to change from its initial value (40%) to some chosen value (say, 20%). Your estimate should be approximately 10 minutes, with proportional control only.

With derivative (as in this trial), estimate the time required for the controller output signal to change from its initial value (40%) to the same point (20%). Your estimate should be approximately five minutes.

Conclusion: With a derivative time of 5 minutes, the controller output signal should lead (or get to the same chosen point) by 5 minutes its behavior with proportional control only.)

Repeat this part of this laboratory exercise, this time with a measurement ramp rate or a derivative time of your choice. You should observe:

Use ramped value instead of value from process sensor? ☐ No ☒ Yes

Ramp Rate EU/min

Direction ☐ Down ☐ Stop ☒ Up

Deriv Minutes

$$\text{Derivative mode contribution to controller output} = - \text{Gain} \times \text{Deriv time} \times \text{Rate of change of measurement}$$

$$\text{Derivative mode contribution} = -0.5 \times 3 \times 3\% = -4.5\%$$

The “-” sign is due to the fact that the derivative contribution is always in a direction which **opposes** the direction of measurement change.

